SIMATS ENGINEERING
(SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

CO4 ASSESSMENT TOOL 1

Test Case Design Assignment

Course Code:	CSA1017	**Course Title:**	Software Engineering
CO Assessed:	CO4	**Assessment:**	Assessment Tool 1 – Test Case Design Assignment
Weightage:	20%	**Total Marks:**	25

CO4 Statement: Implement robust testing, quality assurance, and DevOps practices, emphasizing security, reliability, and ethics

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Problem Statement No.:	39	**Duration:**	60 minutes
Assigned Problem Statement:	Intelligent AI-Based Airport Baggage Tracking and Management System		

Code of Conduct

I, **RR Thiyagesh** (Reg. No.: **192572007**), certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ______________________________

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1. Problem Statement Overview

Assigned Problem (39): Intelligent AI-Based Airport Baggage Tracking and Management System

Airports handle millions of passenger bags annually, and efficient baggage handling is essential for operational performance and passenger satisfaction. Baggage delays, loss, and misrouting increase operational costs and cause passenger inconvenience. The proposed AI-powered platform uses RFID, IoT, cloud, and AI technologies to track baggage in real time, optimize routing, predict delays, notify passengers automatically, and generate operational analytics, thereby improving baggage handling efficiency and reducing complaints.

2. Functional and Non-Functional Requirements

2.1 Functional Requirements (FR)

- FR1: The system shall register and tag baggage with a unique RFID identifier at check-in.
- FR2: The system shall track baggage location in real time at each checkpoint (check-in, sorting, loading, transit, unloading, delivery).
- FR3: The system shall optimize baggage routing across sorting and transfer points.
- FR4: The system shall use AI models to predict potential baggage delays or misrouting risk.
- FR5: The system shall send automated notifications to passengers on baggage status changes.
- FR6: The system shall flag and log misrouted or lost baggage for corrective action.
- FR7: The system shall generate operational analytics and reports for airport/airline staff.
- FR8: The system shall allow authorized staff to search and query the status of any bag by tag ID.

2.2 Non-Functional Requirements (NFR)

- NFR1 (Performance): Baggage location updates shall be reflected within 5 seconds of an RFID scan.
- NFR2 (Reliability): The system shall maintain 99.5% uptime during airport operating hours.
- NFR3 (Scalability): The platform shall support tracking of at least 50,000 bags per day per hub.
- NFR4 (Security): Passenger and baggage data shall be encrypted in transit and at rest; access shall be role-based.
- NFR5 (Usability): The passenger notification interface shall be understandable without training.
- NFR6 (Accuracy): AI delay/misroute predictions shall achieve at least 90% precision on historical validation data.
- NFR7 (Interoperability): The system shall integrate with existing airline and airport handling systems via standard APIs.

2.3 Scope and Assumptions

Testing focuses on the baggage-tracking, routing, prediction, notification, and analytics functions described in the problem statement. It assumes that RFID hardware, network connectivity, and the AI prediction model are available as external dependencies and are already integrated with the platform under test; hardware-level RFID signal testing and the internal training of the AI model itself are outside the scope of this document. Passenger contact details and flight-connection data are assumed to be supplied correctly by upstream airline systems unless a test case explicitly targets that data as invalid.

3. System Architecture

The architecture below shows how the layers interact — data capture at the gate, sorting and handling interfaces feed into a cloud platform and AI predictive engine, which in turn drive passenger-facing and operational applications. This forms the basis for scoping which components each test case exercises.

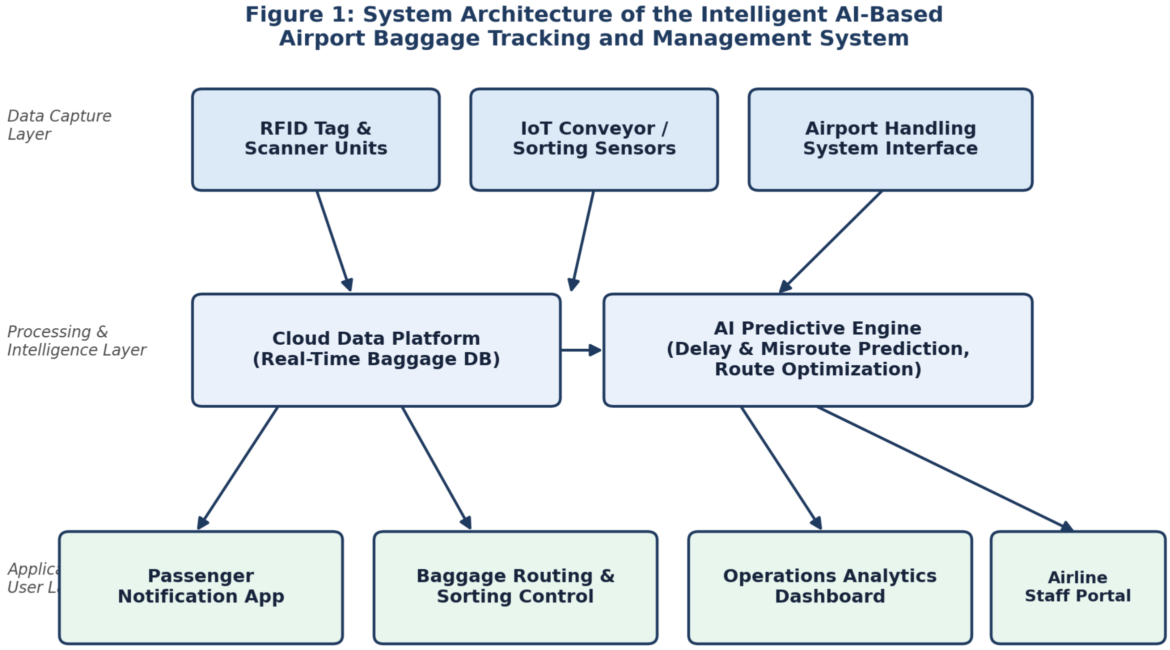

Figure 1: System Architecture of the AI-Based Baggage Tracking Platform

4. Identified Test Scenarios

- TS1: Verify baggage is correctly registered and RFID-tagged at check-in.
- TS2: Verify real-time location updates as baggage passes each checkpoint.
- TS3: Verify baggage routing optimization directs bags to the correct connecting flight/belt.
- TS4: Verify AI delay/misroute prediction triggers appropriate alerts.
- TS5: Verify passenger notifications are sent for key baggage status changes.
- TS6: Verify misrouted/lost baggage is flagged, logged, and escalated correctly.
- TS7: Verify operational analytics reports reflect accurate, up-to-date data.
- TS8: Verify staff search/query of baggage status by tag ID returns correct results.
- TS9: Verify system behaviour under invalid, missing, or duplicate RFID scans.
- TS10: Verify system performance and data integrity under high baggage volume (peak load).

5. Test Design Techniques Applied

5.1 Equivalence Partitioning (EP)

Applied to the RFID tag ID field and baggage weight input. Valid partition: well-formed tag ID (e.g., 12-character alphanumeric) and weight within 0–32 kg. Invalid partitions: malformed/empty tag ID, negative weight, and weight exceeding the airline maximum limit.

5.2 Boundary Value Analysis (BVA)

Applied to baggage weight limits (economy limit = 23 kg, excess-baggage threshold = 32 kg) and to the connection-time window used for misroute-risk prediction (minimum connection time = 45 minutes). Boundary values tested: 22 kg, 23 kg, 24 kg, 31 kg, 32 kg, 33 kg and 44/45/46 minutes connection time.

5.3 Decision Table Testing

Applied to the passenger-notification and misroute-escalation logic, where multiple conditions (RFID scan success, checkpoint match, predicted delay flag, connection time) combine to decide the system's action (notify passenger, reroute bag, escalate to staff).

Rule	RFID Scan Successful	Checkpoint Matches Route	Predicted Delay Risk	System Action
R1	Yes	Yes	No	Update status; no alert
R2	Yes	No	No	Flag as misrouted; alert staff
R3	Yes	Yes	Yes	Notify passenger of possible delay
R4	No	–	–	Log scan failure; request rescan
R5	Yes	No	Yes	Reroute bag; notify passenger and staff

5.4 State Transition Testing

Applied to the baggage lifecycle, which moves through a defined sequence of states from check-in to delivery, with exception branches for misrouting and loss. Test cases verify both valid transitions and invalid/unexpected transitions (e.g., attempting to mark a bag ‘Delivered’ without it having been ‘Unloaded at Destination’).

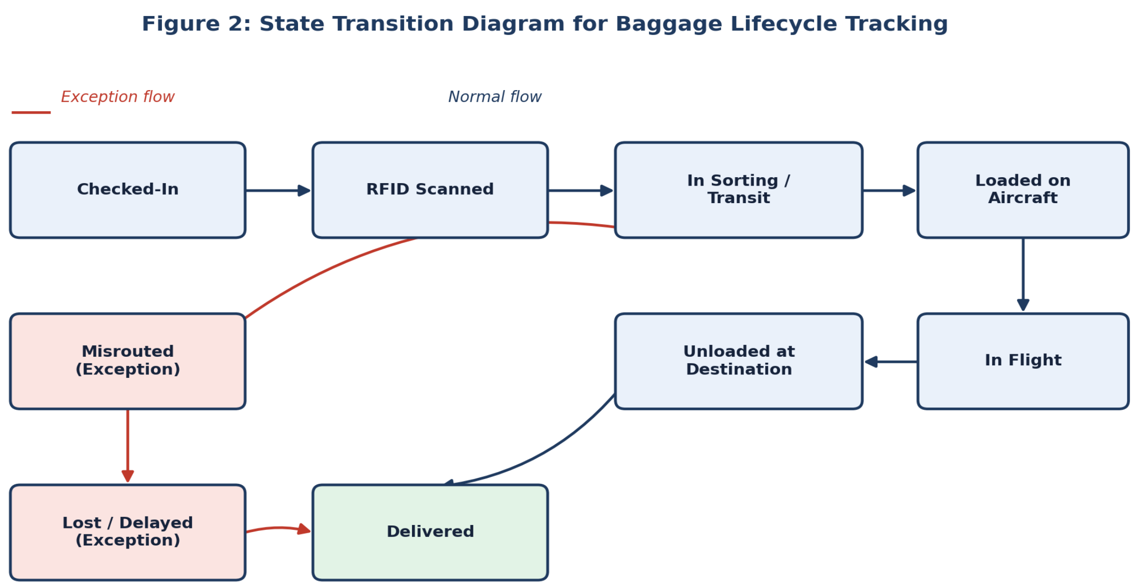

Figure 2: State Transition Diagram for Baggage Lifecycle Tracking

6. Detailed Test Cases

The following test cases cover normal, boundary, invalid, and exceptional conditions across the core functional requirements, with test data and precise expected results. The Actual Result and Pass/Fail Status columns are left for completion during execution.

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Pass/ Fail
TC_01	Normal: Baggage registration and RFID tagging (FR1)	1. Passenger checks in bag. 2. Staff prints and attaches RFID tag. 3. System scans tag.	Valid 12-char alphanumeric tag ID; weight = 18 kg	Bag is registered with unique tag ID; status set to 'Checked-In'.		
TC_02	Normal: Real-time checkpoint tracking (FR2)	1. Bag passes RFID scanner at sorting point. 2. System records checkpoint event.	Tag ID from TC_01; checkpoint = 'Sorting Area 2'	Status updates to 'In Sorting/Transit' within 5 seconds (NFR1).		
TC_03	Boundary: Baggage weight at economy limit (BVA, FR1)	1. Enter baggage weight during check-in.	Weight = 23 kg (boundary)	Bag accepted at standard fare; no excess-baggage flag raised.		
TC_04	Boundary: Baggage weight just above economy limit (BVA)	1. Enter baggage weight during check-in.	Weight = 24 kg	System raises excess-baggage notice; fare recalculated.		
TC_05	Boundary: Baggage weight at maximum allowed limit (BVA)	1. Enter baggage weight during check-in.	Weight = 32 kg (boundary)	Bag accepted with excess-baggage charge applied.		
TC_06	Invalid: Baggage weight exceeds maximum limit (EP)	1. Enter baggage weight during check-in.	Weight = 33 kg	System rejects check-in; displays 'Exceeds maximum allowed weight' error.		
TC_07	Invalid: Negative or zero baggage weight (EP)	1. Enter baggage weight during check-in.	Weight = -2 kg / 0 kg	System rejects input; displays validation error, no bag record created.		
TC_08	Invalid: Malformed/empty RFID tag ID (EP, FR1)	1. Attempt to register bag with invalid tag format.	Tag ID = '' (empty) / '12A' (too short)	System rejects registration; prompts staff to reprint a valid tag.		
TC_09	Normal: Baggage routing optimization (FR3)	1. Bag with connecting flight is scanned at transfer point. 2. System computes optimal route.	Connection time = 90 minutes; 2 valid transfer belts	System assigns bag to the fastest available sorting route to the connecting flight.		
TC_10	Boundary: Minimum connection time threshold (BVA, FR4)	1. Bag arrives with tight connection. 2. AI engine evaluates misroute/delay risk.	Connection time = 45 minutes (boundary)	System marks connection as 'at risk' and prioritizes routing, per threshold rule.		
TC_11	Boundary: Connection time just below minimum threshold (BVA)	1. Bag arrives with very tight connection.	Connection time = 44 minutes	System flags high misroute risk and triggers proactive passenger notification (FR5).		
TC_12	Normal: AI delay prediction and alert (FR4)	1. Historical + live data fed into AI model. 2. Model predicts delay probability.	Simulated delay probability = 0.85 (high)	System raises a delay-risk alert to operations staff with predicted delay window.		
TC_13	Normal: Passenger notification on status change (FR5)	1. Baggage status changes to 'Loaded on Aircraft'. 2. Notification service triggers.	Valid passenger contact (email/SMS) linked to tag ID	Passenger receives notification within 5 seconds of the status change.		
TC_14	Exceptional: Notification with unreachable passenger contact (FR5)	1. Trigger status-change notification. 2. Passenger contact is invalid/unreachable.	Malformed email address on file	System logs delivery failure and retries; does not crash notification service.		
TC_15	Exceptional: Misrouted baggage detection and escalation (FR6)	1. Bag scanned at checkpoint inconsistent with its assigned route.	Checkpoint scan does not match expected route ID	System flags bag as 'Misrouted', logs event, and alerts baggage-handling staff.		

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Pass/ Fail
TC_16	Exceptional: Lost baggage (no scan for extended period) (FR6)	1. No RFID scan recorded for a bag beyond the expected time window.	No scan event for > 4 hours after last checkpoint	System auto-flags bag as 'Lost/Delayed' and initiates the lost-baggage workflow.		
TC_17	State Transition – Invalid: Skip-state transition attempt	1. Attempt to set status directly from 'Checked-In' to 'Delivered'.	Direct status update request bypassing intermediate states	System rejects the invalid transition and retains the last valid state.		
TC_18	Normal: Operational analytics report generation (FR7)	1. Staff requests daily baggage-handling report.	Report date range = current day	System generates report with accurate counts of on-time, delayed, and misrouted bags.		
TC_19	Normal: Staff search/query by tag ID (FR8)	1. Staff enters tag ID in search interface.	Valid existing tag ID	System returns current status, location, and history for that bag within 2 seconds.		
TC_20	Invalid: Search with non-existent tag ID (EP, FR8)	1. Staff enters a tag ID not present in the system.	Tag ID = 'ZZ0000000000' (not registered)	System displays 'No record found' message; no application error.		
TC_21	Exceptional: Duplicate RFID scan at same checkpoint	1. Same tag ID scanned twice in quick succession at one checkpoint.	Two scans of the same tag ID within 2 seconds	System deduplicates the event and records only one checkpoint entry.		
TC_22	Non-functional: Peak-load performance (NFR2, NFR3)	1. Simulate 50,000 bag-scan events within a single operating day.	50,000 concurrent/near-concurrent scan events	System maintains 99.5% uptime and processes updates within the 5-second target.		
TC_23	Non-functional: Data security and access control (NFR4)	1. Unauthorized user attempts to access baggage/passenger data.	Login with staff account lacking baggage-data role	Access is denied; attempt is logged; data remains encrypted in transit and at rest.		
TC_24	Non-functional: AI prediction accuracy validation (NFR6)	1. Run AI model against labelled historical delay/misroute dataset.	Historical validation dataset (≥ 1000 labelled records)	Model achieves ≥ 90% precision on delay/misroute prediction.		

7. Test Case Design and Traceability Process

The diagram below summarizes the end-to-end process used to derive the test cases in Section 6 from the requirements in Section 2, including the feedback loop used once test cases are executed and actual results are compared against expected results.

Figure 3: Test Case Design and Traceability Process Flow

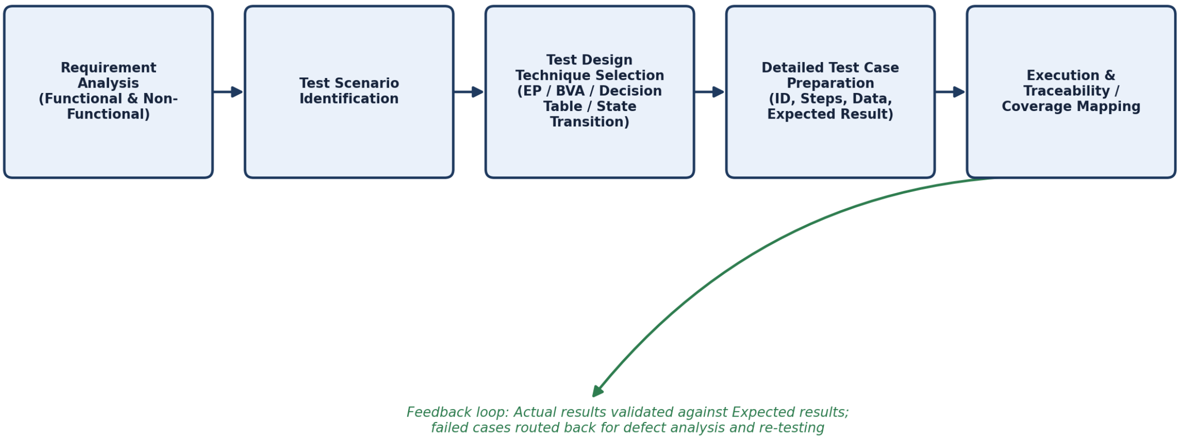

Figure 3: Test Case Design and Traceability Process Flow

8. Requirement Traceability Matrix

Requirement	Test Scenario(s)	Test Case ID(s)
FR1 – Baggage registration & tagging	TS1, TS9	TC_01, TC_08
FR2 – Real-time checkpoint tracking	TS2	TC_02
FR3 – Baggage routing optimization	TS3	TC_09
FR4 – AI delay/misroute prediction	TS4	TC_10, TC_11, TC_12
FR5 – Passenger notifications	TS5	TC_13, TC_14
FR6 – Misrouted/lost baggage handling	TS6	TC_15, TC_16, TC_17
FR7 – Operational analytics reports	TS7	TC_18
FR8 – Staff search/query by tag ID	TS8	TC_19, TC_20
NFR1/NFR2/NFR3 – Performance, reliability, scalability	TS10	TC_22
NFR4 – Security and access control	–	TC_23
NFR6 – AI prediction accuracy	–	TC_24
General – Invalid/duplicate data handling	TS9	TC_06, TC_07, TC_21

10. Test Environment and Entry/Exit Criteria

10.1 Test Environment

• Staging deployment of the baggage-tracking platform (cloud data platform, AI predictive engine, notification service, analytics dashboard) isolated from live airport operations.

• Simulated RFID scanner and IoT sorting-sensor feeds used to generate checkpoint events, including valid, malformed, and duplicate scan events.

• Test passenger and flight-connection dataset covering normal, boundary, and invalid weight, connection-time, and contact-detail values.

• Historical, labelled delay/misroute dataset for validating AI prediction accuracy (NFR6).

• Role-based staff accounts of varying privilege levels for access-control testing (NFR4).

10.2 Entry Criteria

• Functional and non-functional requirements (Section 2) are reviewed and approved.

• Test environment and simulated RFID/IoT data feeds are configured and available.

• Test cases in Section 6 are reviewed and test data is prepared.

10.3 Exit Criteria

• All planned test cases (TC_01–TC_24) have been executed at least once.

• No open critical or high-severity defects remain in baggage tracking, routing, or notification functions.

• Requirement traceability coverage (Section 8) is complete, with every FR and tested NFR mapped to at least one executed test case.

• AI prediction accuracy and system performance meet the thresholds defined in NFR6 and NFR1–NFR3.

11. Test Coverage Summary

Condition Type	Test Design Technique(s) Used	Test Case Count
Normal (valid, expected flow)	Equivalence Partitioning, State Transition	8
Boundary	Boundary Value Analysis	4
Invalid	Equivalence Partitioning	5
Exceptional	State Transition, Decision Table	4
Non-functional (performance, security, accuracy)	N/A – requirement-based scenarios	3

In total, 24 detailed test cases were designed against the 8 functional and 7 non-functional requirements identified for the Intelligent AI-Based Airport Baggage Tracking and Management System, covering normal, boundary, invalid, and exceptional conditions, with Equivalence Partitioning, Boundary Value Analysis, Decision Table Testing, and State Transition Testing applied wherever appropriate.

12. Conclusion

The test cases documented in this assignment provide adequate requirement and functional coverage for the assigned baggage tracking and management system. By combining Equivalence Partitioning, Boundary Value Analysis, Decision Table Testing, and State Transition Testing, the test suite verifies correctness under normal operating conditions while also validating the system's behaviour at input boundaries, with invalid data, and during exceptional events such as misrouted or lost baggage. The Requirement Traceability Matrix in Section 8 confirms that every functional and non-functional requirement identified in Section 2 is exercised by at least one test case, supporting confidence in the reliability, security, and overall quality of the proposed platform ahead of execution and defect logging.